



Policy–Technology Integration and Sustainable Urban Mobility: Evidence from Metropolitan Transport Systems in Indonesia

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Abstract: Urbanisation across Indonesia’s metropolitan regions has intensified pressure on transport systems, manifesting in persistent congestion, environmental degradation, and structural dependence on private vehicles. Addressing these challenges requires coordinated alignment between transport policy frameworks and the deployment of emerging mobility technologies. This study investigates how policy–technology integration shapes sustainable public transport use within metropolitan transport systems, with particular attention to the role of Urban Density in conditioning behavioural responses. A cross-sectional dataset was collected from 500 public transport users across eleven officially designated metropolitan regions in Indonesia. Structural relationships among key constructs were examined using partial least squares structural equation modelling (PLS-SEM). The analysis demonstrates that both transport policy instruments and digital mobility adoption exert significant influences on perceived service quality and user disposition towards public transport. Among these factors, perceived service quality emerges as the most direct determinant of sustained usage behaviour. In addition, Urban Density is found to significantly moderate the linkage between user disposition and actual behaviour, indicating that high-density metropolitan contexts strengthen the translation of preferences into consistent transport choices. The findings highlight the importance of integrating regulatory measures with digital mobility infrastructures to improve system-level performance and user experience in public transport networks. From a policy perspective, the study underscores the need for metropolitan authorities to adopt coordinated governance strategies that align technological deployment with service provision and spatial planning conditions. These insights contribute to ongoing discussions on sustainable urban mobility by situating behavioural outcomes within a broader transport system and policy integration framework.

Keywords: Behavioral change; Indonesia; Metropolitan governance; PLS-SEM; Public attitude; Service quality; Sustainable transport policy; Technology adoption

1 Introduction

Urban transport has become one of the most pressing global challenges of the 21st century, closely linked to rapid urbanisation, increasing congestion, rising carbon emissions, and declining quality of life in cities. It is estimated that by 2050, more than two-thirds of the world’s population will live in urban areas, with Asia experiencing the sharpest growth, from the current urbanisation rate of 48% to more than 64% in the coming decades [1, 2]. This unprecedented demographic shift has put pressure on urban systems, particularly transportation infrastructure, where demand far exceeds available capacity. In fast-growing cities, motorisation is on the rise, driving high private vehicle ownership and worsening traffic congestion [3]. The resulting congestion not only delays mobility but also erodes economic productivity, causing billions of dollars in losses each year [4]. Another worrying transport parallel issue is the surge in carbon emissions, as urban transport remains one of the most significant contributors to greenhouse gases, especially CO₂ [5]. Dependence on motor vehicles causes inefficient energy consumption. The impacts include respiratory diseases, stress, and decreased life expectancy, which ultimately weaken metropolitan housing and sustainability.

This situation is most pronounced in Southeast Asia, where megacities such as Jakarta, Manila, and Bangkok reflect the multidimensional challenges of urbanisation. The region is facing rapid demographic expansion, requiring around 50 million new housing units by 2030, while grappling with limited transportation infrastructure, inadequate public transportation coverage, and worsening congestion [1]. These structural limitations are exacerbated by environmental vulnerabilities, including floods and heat waves, that threaten already stressed systems [6]. Social inequality further complicates the situation, as the growth of informal settlements and slums—home to tens of millions of people—limits equitable access to reliable mobility and decent living conditions [7].

The growth of metropolitan areas in Indonesia, a developing country with one of the highest levels of congestion in the world, increasingly displays complex dynamics influenced by rapid urbanisation, increased population mobility, and limited transportation infrastructure. Demographic trends show that the proportion of the population living in urban areas has surpassed 58% in 2023 and is expected to increase to nearly 60% by 2025, with major cities such as Jakarta, Surabaya, and Bandung becoming the epicentres of urban concentration [8, 9]. Extreme density, exemplified by the Bandung area with more than 15,000 people per km² and Central Jakarta as the most populous area in the capital, emphasises the pressure on the transportation system [10]. The impact of this condition is most evident in increasingly congested traffic. Bandung is now recorded as the city with the highest level of congestion in Indonesia, with an average travel time of 32 minutes and 37 seconds per 10 km and a density rate of 48%. Surabaya and Jakarta also occupy the top five positions, although Jakarta has made efforts to develop public transportation modes such as TransJakarta and light-rail transit (LRT) [8].

Dependence on private vehicles is the main feature of urban transportation in Indonesia's major cities. The growth of motorcycle and private car ownership continues to increase due to the flexibility offered, while public transportation options are still limited and not yet fully integrated [11]. This creates a cycle of problems: the more private vehicles on the road, the worse the congestion becomes, especially during rush hour in business centres and densely populated residential areas. Public transportation infrastructure, such as rapid transit buses, commuter trains, and LRT, still faces challenges in the form of limited coverage, capacity, and intermodal integration, so it has not been able to become an effective alternative for the majority of the population [12]. As a result, air pollution has increased significantly, with the concentration of fine particulate matter (PM_{2.5}) in Jakarta often exceeding WHO standards. In addition, the economic costs due to congestion are enormous, characterised by loss of work productivity, waste of fuel, and increasing health costs [13]. These issues underscore the need for comprehensive transportation reform.

Sustainable transportation is now seen as a key pillar in building an inclusive, resilient, and environmentally friendly city. This concept includes various strategies such as the development of mass public transportation, non-motorised mobility (walking and cycling), the adoption of electric vehicles (EVs), and the use of intelligent mobility based on digital technology [14–16]. An integrated mass public transport system plays a vital role in reducing congestion, reducing emissions, and providing safe and affordable mobility access for all levels of society, in line with SDG 11's target of sustainable cities [17, 18]. Non-motorised mobility, such as walking and cycling, also supports healthier, greener cities and reduces dependence on motor vehicles [19]. On the other hand, the adoption of EVs presents a great opportunity to lower carbon emissions and air pollution, although it requires clear infrastructure and policy investments to accelerate their penetration [20, 21]. The support of Internet of Things (IoT)-based smart mobility technology and data analytics further improves the efficiency, safety, and resilience of the transportation system [22]. Sustainable transport is an essential bridge between SDGs 11 and SDG 13, not only supporting urban affordability and inclusion, but also driving global climate action through reducing emissions and strengthening urban resilience [23, 24].

Various recent empirical studies confirm that transportation policy instruments, including regulations, subsidies, incentives, and disincentives, can significantly change the behavior of transportation users. For example, congestion pricing policies have been proven to encourage the switch from private vehicles to more sustainable modes of public transportation [25]. Regulations on ride-hailing services also show a positive impact in reducing emissions by limiting the number of vehicles and reducing environmental externalities [26]. On the other hand, subsidy and incentive policies play an essential role in improving accessibility and quality of services. The fare-free public transport program can increase the number of passengers, especially from vulnerable groups, although it requires careful planning to avoid overcapacity and fiscal burdens [27, 28]. Subsidies that increase access and comfort of public transportation are also effective in encouraging the shift from private vehicles [29]. On the other hand, disincentives in the form of fuel tax increases or parking fees have been proven to reduce reliance on private cars while encouraging active mobility options such as walking and cycling [30]. Theoretically, the theory of planned behavior (TPB) and Random Utility Theory approaches show that policies that touch on the rational and psychological aspects of users tend to be more effective in shaping transportation behavior [31, 32].

In addition to policies, the adoption of digital transportation technology has been a major catalyst in shifting user behavior. Ride-hailing apps have been proven to change travel patterns by offering high flexibility and comfort, influenced by usability, ease of use, and social influence [33, 34]. E-ticketing technology also strengthens efficiency, reduces transaction costs, and increases user satisfaction, thereby accelerating the transition to digital-based public

modes [35, 36]. Furthermore, multi-sided digital platforms such as Uber affirm the role of gamification and added value in driving user loyalty as well as disrupting traditional transportation services [37]. The integration of big data and e-mobility enables the personalisation of services while improving system efficiency through user preference mapping [38, 39]. Thus, the combination of the right transport policy and the adoption of innovative technologies is not only directing mobility behavior towards a more sustainable pattern, but also accelerating the transition of cities towards an innovative, low-emission transport ecosystem.

However, the existing empirical literature on urban transportation has tended to separate the focus between policy effectiveness and technology adoption. While behavioral theories such as the TPB and technology acceptance frameworks, e.g., Technology Acceptance Model (TAM) and Unified Theory of Acceptance and Use of Technology (UTAUT), have been widely applied to explain mobility choices, these approaches primarily focus on individual cognitive determinants such as attitudes, perceived usefulness, subjective norms, or behavioral intentions. Although these models provide valuable insights into individual decision-making processes, they often overlook the broader structural context in which mobility behavior occurs. In metropolitan transport systems, individual travel behavior is not shaped solely by psychological preferences but also by institutional interventions such as transport regulations, pricing mechanisms, subsidies, and infrastructure policies, as well as by the rapid diffusion of digital mobility technologies, including ride-hailing platforms, e-ticketing systems, and smart mobility applications.

Despite the growing body of research on sustainable mobility, most empirical studies continue to examine transport policy instruments and technology adoption separately. Policy-oriented studies typically focus on regulatory effectiveness or pricing mechanisms, whereas technology-focused research emphasises digital platforms and user acceptance of mobility innovations. This separation limits our understanding of how governance-driven policies and technological systems interact to influence behavioral change toward sustainable mobility. By integrating transport policy instruments and technology adoption within a single behavioral framework, this study advances existing models by linking structural policy interventions and digital mobility technologies with behavioral outcomes through the mediating roles of public attitudes and perceived service quality. This integrated perspective provides a more comprehensive explanation of sustainable mobility behavior in metropolitan contexts, particularly in rapidly urbanising developing countries such as Indonesia.

Based on the practical gaps and existing empirical evidence, three research questions arise:

RQ1. How do sustainable urban transport policies and Technology Adoption in Urban Transport influence public attitudes toward sustainable mobility and perceived service quality of public transport in Indonesian metropolitan areas?

RQ2. To what extent do public attitudes toward sustainable mobility and perceived service quality of public transport affect actual sustainable public transport use behavior?

RQ3. How does Urban Density moderate the relationship between public attitudes, perceived service quality, and actual sustainable public transport use behavior?

The primary objective of this study is to investigate the integrated influence of sustainable urban transport policies and technology adoption on public attitudes, perceived service quality, and actual sustainable public transport use behavior in Indonesian metropolitan areas. Specifically, the study aims to assess how governance-driven policies and digital innovations interact to shape behavioral change, and how Urban Density moderates these relationships. This study makes several theoretical contributions. First, it advances the sustainable mobility literature by integrating transport policy instruments and digital mobility technology adoption within a unified behavioral framework. Second, it extends existing behavioral models by demonstrating how structural governance interventions and technological systems jointly shape public attitudes, perceived service quality, and sustainable transport behavior. Third, by incorporating Urban Density as a moderating factor, the study highlights how metropolitan spatial characteristics condition the effectiveness of policy and technology interventions.

2 Literature Review

2.1 Sustainable Urban Transport Policy and Metropolitan Governance

Rapid urbanisation, rising motorisation, and increasing environmental pressures have made sustainable urban transport a central challenge for metropolitan regions worldwide. In response, scholars increasingly emphasise the importance of integrating sustainable transport policy with effective metropolitan governance to create mobility systems that are environmentally responsible, socially inclusive, and economically efficient [40]. Transport policy interventions, such as investments in public transport, land-use integration, and smart mobility technologies, are widely recognised as key mechanisms for shaping sustainable urban mobility [41].

Public transportation remains the cornerstone of sustainable mobility systems. Studies consistently show that investments in transport infrastructure, fleet electrification, and operational efficiency improvements can significantly enhance the environmental and operational sustainability of urban transport networks [42]. Similarly, the integration of transport planning with land-use development—particularly through strategies such as compact urban design and transit-oriented development (TOD)—has been shown to reduce dependence on private vehicles while improving

accessibility and travel efficiency [43]. In parallel, digital innovations and smart mobility solutions are increasingly viewed as enabling tools for improving transport system performance. Technologies such as intelligent transportation systems (ITS), mobility data platforms, and digitally integrated ticketing systems contribute to reducing congestion, improving operational efficiency, and supporting transport decarbonisation [44, 45]. In addition, active mobility options such as walking and cycling are increasingly recognised as critical components of sustainable transport systems due to their benefits for environmental sustainability, public health, and social inclusion [46, 47].

While these policy instruments are widely discussed in the literature, their effectiveness often depends on the quality of metropolitan governance arrangements that coordinate transport planning across jurisdictions and institutional actors. Metropolitan areas typically involve multiple administrative units, making transport governance highly complex. Previous studies highlight that collaborative governance frameworks involving government, private sector actors, and communities are essential for implementing effective urban transport policies [48]. However, institutional fragmentation frequently undermines policy implementation, as transport responsibilities are distributed across different levels of government and agencies [49]. To address this challenge, scholars emphasise the need for integrated planning frameworks and multi-level governance mechanisms capable of coordinating policies across sectors and jurisdictions [19].

Despite these advancements, several persistent barriers continue to constrain the effectiveness of sustainable urban transport governance. Governance fragmentation, limited financial and technical resources, and weak public participation often reduce the legitimacy and implementation capacity of transport policies, particularly in developing countries [50]. These challenges highlight the importance of institutional structures capable of bridging national and local policy agendas while fostering cross-sectoral collaboration and stakeholder engagement [51]. Policy instruments such as TOD development, parking management, and improvements in public transport services therefore require governance systems that balance environmental sustainability, economic feasibility, and social equity [52, 53].

The literature broadly agrees that sustainable urban transport requires an integrated combination of policy instruments, infrastructure investment, and effective metropolitan governance coordination. However, existing research has largely examined these elements from a policy or governance perspective without systematically analysing how such institutional frameworks influence public behavioral responses to sustainable transport initiatives. Consequently, the behavioral implications of governance-driven transport policies remain insufficiently explored.

2.2 Technology Adoption in Urban Transportation Management

The rapid development of digital technologies has significantly transformed urban transportation management. Technology adoption in this sector is increasingly viewed as a strategic mechanism to improve operational efficiency, reduce congestion, lower carbon emissions, and enhance user experience [22, 54]. Recent literature emphasises several key technological innovations—such as ITS, data-driven mobility management, and integrated digital mobility platforms—that are reshaping how urban transport systems operate.

ITS represents one of the most widely implemented technological innovations in urban mobility. These systems include real-time traffic information services, adaptive traffic signal control, and smart parking management, which collectively improve traffic flow and reduce travel delays [55, 56]. The expansion of IoT infrastructure and big data analytics further strengthens this transformation by enabling cities to monitor traffic conditions and analyse mobility patterns through data generated from sensors, GPS-enabled vehicles, and ride-hailing platforms [57]. Such data-driven approaches allow policymakers to design evidence-based transport strategies and optimise system performance, although challenges related to data privacy, cybersecurity, and governance remain significant [58].

Another important dimension of technological innovation is the emergence of integrated digital mobility platforms, particularly through the concept of Mobility-as-a-Service (MaaS). MaaS integrates multiple transportation modes—such as public transport, ride-hailing services, shared bicycles, and micro-mobility—within a single digital platform, enabling users to plan, book, and pay for trips seamlessly [59, 60]. Empirical studies suggest that such integration can enhance travel convenience, improve multimodal connectivity, and potentially reduce dependence on private vehicles [61]. Similarly, digital payment systems and e-ticketing solutions have been shown to simplify transactions and improve service accessibility in public transportation systems. Despite these benefits, the adoption of digital transportation technologies remains uneven across cities. Existing research highlights several enabling factors, including supportive government policies, infrastructure investment, and increasing public awareness of environmental sustainability [62, 63]. At the same time, technological adoption can be constrained by financial limitations, institutional fragmentation, user resistance to new systems, and the lack of coordination between public and private stakeholders [64]. In addition, social and technological inequalities—often referred to as the digital divide—may limit access to digital mobility services in areas with low technology literacy or uneven digital infrastructure [65].

The existing literature consistently recognises the potential of digital technologies to improve the efficiency, sustainability, and inclusiveness of urban transportation systems. However, most studies examine technology adoption primarily from a technological or operational perspective, focusing on system performance or user acceptance.

Less attention has been given to how technology adoption interacts with broader transport policy frameworks and governance structures to influence behavioral change in urban mobility.

2.3 Underpinning Theory: Theory of Planned Behavior in Transportation Studies

The TPB, proposed by Ajzen [66, 67], is one of the most widely used behavioral frameworks for explaining individual decision-making processes. The theory posits that human behavior is primarily determined by behavioral intention, which in turn is shaped by three key factors: attitude toward the behavior, subjective norms, and perceived behavioral control (PBC). Attitudes refer to an individual's positive or negative evaluation of a behavior, subjective norms reflect perceived social pressures to perform or avoid a behavior, and PBC captures an individual's perception of their ability to perform the behavior. Together, these factors influence behavioral intentions, which subsequently shape actual behavior.

In transportation research, TPB has been widely applied to explain travel mode choices and sustainable mobility behavior. Previous studies consistently find that positive attitudes toward public transportation or environmentally friendly mobility significantly increase individuals' intentions to adopt sustainable transport modes [68, 69]. Social influences also play an important role, as individuals' travel decisions are often shaped by perceived expectations and behavioral patterns within their social environment. Similarly, PBC—such as perceived convenience, accessibility, or ability to switch transport modes—has been shown to influence both behavioral intention and actual travel behavior [70, 71].

Beyond traditional travel behavior studies, TPB has also been applied to examine the adoption of emerging mobility options such as EVs, cycling, and other sustainable transport alternatives. These studies generally confirm that attitudes, social norms, and PBC remain key psychological determinants influencing individuals' willingness to adopt sustainable mobility practices [72, 73]. In recent years, scholars have further extended TPB by incorporating additional factors—such as environmental awareness, moral norms, and habitual behavior—to better capture the complexity of sustainable transport decisions [74, 75].

Despite its strong explanatory power, TPB-based studies in transportation have primarily focused on individual psychological determinants of travel behavior, while giving relatively limited attention to the broader structural factors that shape mobility decisions. In urban transportation systems, behavioral choices are not only influenced by attitudes or social norms but also by institutional policies, transport infrastructure, and technological innovations that determine mobility opportunities and service quality. Consequently, relying solely on psychological constructs may provide an incomplete explanation of sustainable mobility behavior.

Building on this limitation, the present study extends the TPB perspective by integrating transport policy interventions and technology adoption as contextual drivers that influence behavioral outcomes through public attitudes and perceived service quality. By situating individual behavioral responses within a broader policy and technological environment, this study provides a more comprehensive framework for understanding sustainable urban mobility behavior in metropolitan contexts.

3 Hypothesis Development

3.1 The Influence of Sustainable Urban Transport Policy on Public Attitude Toward Sustainable Mobility

Sustainable Urban Transport Policy refers to a set of regulatory, planning, and investment strategies designed to create a transportation system that meets current mobility needs while preserving environmental, economic, and social sustainability [76]. Such policies typically include investments in public transport infrastructure, promotion of non-motorised mobility, integration of land use and transport planning, and incentives for environmentally friendly transportation technologies [77, 78]. These policy instruments not only reshape the physical structure of urban mobility systems but also influence how individuals perceive and evaluate different transport options.

From a behavioral perspective, public policies can act as contextual drivers that shape individual attitudes toward mobility behavior. According to the TPB, attitudes toward a behavior are formed through individuals' evaluation of the expected outcomes and benefits associated with that behavior [66, 79]. Transport policies that improve service quality, increase accessibility, or enhance environmental benefits can therefore influence how individuals evaluate sustainable mobility options such as public transportation, cycling, or walking. When public transport becomes more reliable, comfortable, and accessible, individuals are more likely to develop favourable perceptions of sustainable transport alternatives.

Sustainable transport policies may influence public attitudes through several mechanisms. First, improvements in public transport service quality—such as higher service frequency, reduced travel time, and greater reliability—can enhance users' perceptions of convenience and efficiency, thereby encouraging more positive evaluations of public transport systems [80, 81]. Second, policies promoting active mobility and environmentally friendly transport options can increase public awareness of the environmental and health benefits associated with sustainable mobility [46, 82]. Third, policy instruments such as information campaigns, incentives, and regulatory measures may influence social norms and public awareness regarding sustainable transportation practices [83, 84].

Empirical studies provide evidence supporting these mechanisms. Previous research shows that improvements in public transport service quality significantly influence individuals' perceptions and attitudes toward sustainable transport modes [85, 86]. Similarly, policy interventions that improve walkability or provide information about sustainable transport options have been shown to increase public support for sustainable mobility initiatives [63, 75]. Economic incentives and structural changes in transport systems may further encourage individuals to reassess the relative benefits of sustainable mobility options compared to private vehicle use [87, 88].

Overall, these theoretical and empirical insights suggest that sustainable urban transport policies can shape individuals' perceptions and evaluations of sustainable mobility, thereby fostering more favourable public attitudes toward environmentally friendly transportation options. Therefore, the following hypothesis is proposed:

Hypothesis 1. Sustainable Urban Transport Policy has a significant effect on Public Attitude Toward Sustainable Mobility.

3.2 The Influence of Sustainable Urban Transport Policy on Perceived Service Quality of Public Transport

Sustainable Urban Transport Policy refers to strategic regulatory and planning interventions aimed at developing transportation systems that are efficient, environmentally friendly, and socially inclusive [89–91]. These policies typically involve investments in public transport infrastructure, fleet modernisation, integration of multimodal transport systems, and improvements in service accessibility. Through these interventions, governments attempt to enhance the performance and attractiveness of public transport systems while reducing dependence on private vehicles and mitigating environmental impacts [89, 90].

From a service management perspective, public policy plays an important role in shaping users' perceptions of service quality by influencing the structural and operational characteristics of transport systems. Service quality in public transportation is commonly evaluated through dimensions such as reliability, safety, comfort, accessibility, and efficiency. When governments implement policies that improve infrastructure, modernise transport fleets, or enhance operational management, these improvements can directly influence how users perceive the quality of services provided.

Sustainable transport policies may affect perceived service quality through several mechanisms. First, investments in infrastructure and system modernisation can improve operational reliability, reduce travel time, and increase service frequency, thereby enhancing perceptions of efficiency and convenience [92]. Second, policies promoting environmentally friendly fleets and integrated transport systems can improve travel comfort and system coordination, which are key elements of perceived service quality. Third, accessibility-oriented policies—such as improving pedestrian connectivity, first-mile and last-mile integration, and inclusive transport design—can increase users' perceptions of service accessibility and usability [93].

Empirical studies support these theoretical arguments. Previous research indicates that reliability, punctuality, comfort, and travel safety are critical determinants of perceived service quality in public transport systems [94, 95]. Studies also highlight the importance of safe and inclusive transport environments, particularly for women and vulnerable groups [85, 96]. In addition, the integration of sustainable mobility services—such as bike-sharing and car-sharing systems—has been shown to enhance perceptions of transport service quality while encouraging a shift away from private vehicle use [86]. Accessibility improvements, particularly those addressing first-mile and last-mile connectivity, further strengthen public perceptions of service convenience and overall quality [97, 98].

Taken together, these theoretical and empirical insights suggest that sustainable urban transport policies can significantly influence how users perceive the quality of public transport services. Policies that enhance system reliability, accessibility, and operational performance are likely to generate more positive evaluations of service quality among transport users.

With a review of the literature and empirical evidence, the following hypotheses can be formulated:

Hypothesis 2. Sustainable Urban Transport Policy has a significant effect on Perceived Service Quality of Public Transport.

3.3 The Influence of Technology Adoption in Urban Transport on Public Attitude Toward Sustainable Mobility

The adoption of digital and intelligent technologies has become a central element in the transformation of urban transportation systems. Technological innovations such as ITS, digital payment platforms, mobility applications, and emerging mobility solutions—including autonomous, connected, electric, and shared (ACES) vehicles—aim to enhance operational efficiency, service convenience, and environmental sustainability in urban mobility systems [99, 100]. These technologies not only improve the functional performance of transportation services but also reshape how users perceive and evaluate sustainable mobility options.

From a behavioral perspective, technology adoption can influence individuals' attitudes toward transportation modes by modifying their perceptions of convenience, usefulness, and overall travel experience. In the context of behavioral theories such as the TPB and technology adoption frameworks, attitudes toward a behavior are influenced

by individuals' evaluations of the expected benefits and outcomes associated with that behavior. When technological innovations make public transport more efficient, reliable, and user-friendly, individuals are more likely to develop favourable attitudes toward sustainable mobility alternatives.

Technology adoption may influence public attitudes through several mechanisms. First, digital innovations such as electronic ticketing, mobile payments, and integrated travel applications simplify the travel process and reduce transaction costs, thereby improving perceived convenience and accessibility of public transport services [101]. Second, intelligent transport technologies—including real-time traffic information systems, route optimisation, and multimodal travel planners—can enhance travel efficiency and reliability, which positively shapes users' perceptions of transport systems [102]. Third, emerging technologies such as shared mobility platforms and electric mobility solutions contribute to environmental sustainability, which may strengthen positive attitudes among individuals with greater environmental awareness [103].

Empirical studies provide evidence supporting these relationships. Previous research shows that users' initial experiences with new mobility technologies, including autonomous vehicles and intelligent transport systems, can significantly influence public perceptions and attitudes toward technology-based mobility solutions [104, 105]. Studies based on the TAM further indicate that perceived usefulness, ease of use, and trust are key determinants influencing individuals' attitudes toward transportation technologies [106, 107]. Similarly, research on MaaS platforms and multimodal travel applications highlights the importance of perceived convenience, efficiency, and environmental benefits in shaping positive attitudes toward sustainable mobility options [43, 108].

Taken together, these theoretical and empirical insights suggest that the adoption of technology in urban transport systems can positively influence public attitudes toward sustainable mobility by improving travel convenience, system efficiency, and environmental performance.

Hypothesis 3. Technology Adoption in Urban Transport has a significant effect on Public Attitude Toward Sustainable Mobility.

3.4 The Influence of Technology Adoption in Urban Transport on Perceived Service Quality of Public Transport

The adoption of digital technologies has become an important driver in improving the performance of public transportation systems. Technological innovations such as smartphone-based mobility applications, ITS, and digital payment platforms aim to enhance the operational efficiency, accessibility, and sustainability of urban transport services [39, 109]. These technologies enable real-time information exchange, integrated ticketing systems, and improved coordination between transport modes, thereby creating more responsive and user-oriented transportation services.

From a service quality perspective, technological innovation can influence users' perceptions of service quality by improving the functional attributes of transportation services. Service quality in public transportation is generally evaluated through dimensions such as reliability, accessibility, punctuality, comfort, and safety [110]. The implementation of digital technologies can enhance these dimensions by improving operational efficiency, reducing uncertainty during travel, and facilitating easier access to transportation services [111]. For instance, real-time information systems and route optimisation technologies allow passengers to make better travel decisions, while automated scheduling systems can improve service reliability and punctuality.

Technological adoption may influence perceived service quality through several mechanisms. First, real-time travel information and smart mobility applications increase transparency and reduce information asymmetry between service providers and users, thereby improving perceived reliability and convenience [112, 113]. Second, intelligent transport technologies can enhance operational efficiency by optimising route planning, scheduling, and traffic management, which directly improves service punctuality and travel time consistency [51, 114]. Third, digital payment systems simplify the ticketing process and reduce transaction barriers, thereby improving accessibility and overall user experience [39].

Empirical evidence supports these theoretical arguments. Previous studies consistently identify service attributes such as travel frequency, punctuality, cost efficiency, and cleanliness as key determinants of perceived service quality and passenger satisfaction [115]. The implementation of real-time tracking systems and automated scheduling technologies has also been shown to improve service reliability and passenger satisfaction in several urban transport systems [116]. In addition, technological innovations can strengthen passengers' perceptions of safety, comfort, and service professionalism, which further contribute to improved perceptions of public transport service quality [117].

Overall, technological innovations in urban transport systems can improve the operational performance and user experience of public transport services, which in turn strengthens users' perceptions of service quality.

With a review of the literature and empirical evidence, the following hypotheses can be formulated:

Hypothesis 4. Technology Adoption in Urban Transport has a significant effect on Perceived Service Quality of Public Transport.

3.5 The Influence of Public Attitude Toward Sustainable Mobility on Actual Sustainable Public Transport Use Behavior

Sustainable mobility is defined as the ability to meet people's needs to move, communicate, trade, and access services without sacrificing ecological and human values in the present and future [78, 118, 119]. The concept emphasises reducing greenhouse gas emissions, pollution, and noise, while improving accessibility, energy efficiency, and transportation safety [120]. Public attitudes towards sustainable mobility are a key factor in the successful implementation of environmentally friendly transportation policies. Indicators such as public transport affordability, multimodal integration, air quality, service reliability, and accident reduction are used to assess mobility sustainability [121, 122]. Public positive attitudes usually arise when public transportation is reliable, safe, and in accordance with people's mobility preferences.

Empirical evidence suggests that public attitudes towards sustainable mobility have a significant influence on sustainable public transport use behavior. A study in Accra, for example, found that attitudes towards public transport actually reduced reliance on private cars, while proximity to city centres and walkability reinforced sustainable mobility behaviors [123]. In addition, psychological factors such as attitudes towards sustainable transportation, PBC, and subjective norms are important determinants in fashion choices. However, altruistic values and environmental attitudes are not always significant drivers, especially in urban contexts with high social complexity [124]. In Delhi, environmental knowledge and personal attitudes are more influential than social norms or government intervention [125]. Users who frequently use shared mobility services are more active in participatory processes, although unequal representation can lead to mobility injustices [126]. Meanwhile, EV acceptance is still fluctuating, influenced by the perception of financial and environmental risks [127, 128].

With a review of the literature and empirical evidence, the following hypotheses can be formulated:

Hypothesis 5. Public Attitude Toward Sustainable Mobility has a significant effect on Actual Sustainable Public Transport Use Behavior.

3.6 The Influence of Perceived Service Quality of Public Transport on Actual Sustainable Public Transport Use Behavior

Perceived Service Quality of Public Transport is defined as a user's subjective perception of the quality of service they receive, encompassing functional, emotional, and evaluative dimensions of their travel experience [129]. This concept is an important indicator in determining user satisfaction and loyalty to public transportation. The main attributes that shape the perception of service quality include functionality and accessibility, namely ease of use, reliability, and availability of transportation modes [52, 98]. Security and safety are also key factors, with positive perceptions of safety measures contributing to perceived quality improvements. [130, 131]. On the other hand, comfort and cleanliness consistently affect satisfaction levels, both in developed and developing countries [132, 133]. In addition, timeliness and reliability are important determinants, where delays or uncertainty of services degrade perceived quality [115, 116]. The availability of information and customer service, including real-time schedule updates and interaction with staff, also strengthens the user experience [129, 130].

Empirical evidence shows that perceived service quality in public transportation has a vital role in shaping the actual behavior of using sustainable modes of transportation. The perception of service quality, which includes aspects of reliability, convenience, security, and ease of access, is consistently associated with user satisfaction and loyalty levels [94, 129]. Dissatisfaction arises when there is a gap between the expected and perceived services, for example, related to punctuality, fares, and driver attitudes. Empirical studies confirm that service quality affects not only satisfaction, but also behavioral intentions to use public transport more often and reduce dependence on private vehicles [86, 134]. In addition, psychological factors such as environmental concern strengthen the relationship between service perception and actual decisions to use public transport [135]. The dimension of perceived accessibility, including ease and comfort of use, plays a vital role as a mediator. Improving the quality of services through security, convenience, and availability of information increases the perception of accessibility and encourages more frequent use [52].

With a review of the literature and empirical evidence, the following hypotheses can be formulated:

Hypothesis 6. Perceived Service Quality of Public Transport has a significant effect on Actual Sustainability of Public Transport Use Behavior.

3.7 The Moderating Role of Urban Density

Urban Density is an important concept in urban planning that refers to the level of concentration of population, buildings, occupations, and human activities in a particular urban area [136]. In measuring Urban Density, several parameters are commonly used. First, population density, which refers to the number of inhabitants per unit area [137]. Second, residential density, which calculates the number of housing units or residences in an area [138]. Third, job density, which is the concentration of economic activity or the number of workers in a particular area [139]. In addition, the floor area ratio (FAR) to the land area is also an important measure to see the level of vertical

space utilisation. Theoretically, Urban Density is closely related to the quality of life and sustainability. Higher levels of density can improve social interaction, improve accessibility of public services, and support transportation efficiency [140, 141]. In the context of sustainability, cities with relatively high density are also considered to be able to reduce urban sprawl and excessive energy consumption [142]. However, too high a density can cause social problems, such as overcrowding, limited open space, and a decrease in comfort in life [143].

Urban Density plays a complex moderating factor in various studies of sustainability and quality of life in cities. From an environmental perspective, high density does not always correlate with a decrease in carbon emissions. The survey in Helsinki showed that high-density regions actually produce greater CO₂ emissions due to higher consumption rates [144]. This is reinforced by findings in South Asia, where population density exacerbates the negative impact of transportation energy consumption on environmental sustainability [145]. In the social aspect, density mediates the quality of life, especially in low-income residential areas. The medium density level is considered the most optimal because it can balance service accessibility, space comfort, and environmental quality, while too high a density tends to trigger overcrowding [146]. In addition, public perception of density is also important; High physical density can reduce social cohesion if it is not balanced with green spaces and public facilities [147]. From a spatial perspective, dense development has the potential to exacerbate the risk of urban flooding because building structures act as sub-watersheds [148]. On the other hand, the compact city approach has also been proven not always to reduce the ecological footprint because socioeconomic factors are more dominant in influencing consumption patterns [149].

With a review of the literature and empirical evidence, this study proposes the following Urban Density moderation hypotheses:

Hypothesis 7. Urban Density moderates the relationship between Public Attitude Toward Sustainable Transport and Actual Sustainability of Public Transport Use Behavior.

Hypothesis 8. Urban Density moderates the relationship between Perceived Service Quality of Public Transport and Actual Sustainability of Public Transport Use Behavior.

Referring to the eight hypotheses that have been formulated, this study presents a model framework as shown in Figure 1.

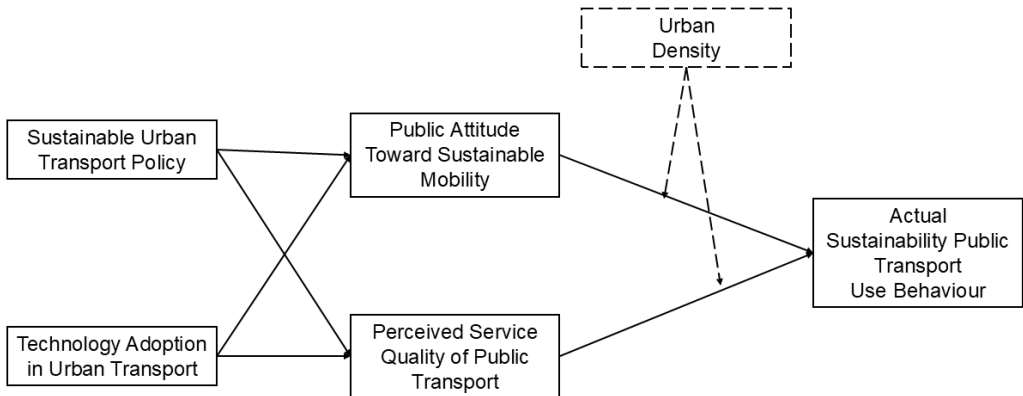


Figure 1. Proposed research model

4 Methodology

4.1 Research Design

This research is designed to explore the combined impact of sustainable urban transport policies and the use of emerging transport technologies on commuter attitudes, perceived service performance, and the actual behavioral shift toward sustainable public transport in Indonesian metropolitan regions. To address this aim, a quantitative cross-sectional survey is employed, as it enables the capture of perceptions and behaviors simultaneously while providing robust data to examine structural associations among key constructs.

The unit of analysis is public transport users in metropolitan regions officially defined by the Indonesian government. These areas, such as Jabodetabekpunjur, Mebidangro, Gerbangkertosusila, and others, represent high-density urban clusters with complex mobility challenges and ongoing policy experiments. Including a wide range of metropolitan contexts enhances the external validity and provides a more comprehensive understanding of governance-related behavioral change. The detailed administrative coverage of each metropolitan region is presented in Table 1.

Table 1. Officially recognised metropolitan areas in Indonesia and their administrative coverage

Metropolitan Area	Main Cities/Districts (Coverage)	Province(s)
Jabodetabekpunjur	Jakarta Special Capital Region, Bekasi City, Bogor City, Depok City, Tangerang City, South Tangerang City, Bekasi Regency, Bogor Regency, Tangerang Regency, part of Cianjut Regency	Jakarta, West Java, Banten
Mebidangro	Medan City, Binjai City, Deli Serdang Regency, Karo Regency	North Sumatra
Patungrava Agung	Palembang City, Banyuasin Regency, Ogan Ilir Regency, Ogan Komering Ilir Regency, Musi Banyuasin Regency	South Sumatra
Bandung Basin	Bandung City, Cimahi City, Bandung Regency, West Bandung Regency, Sumedang Regency	West Java
Kedungsepur	Semarang City, Demak Regency, part of Grobogan Regency, Kendal Regency, Salatiga City, Semarang Regency	Central Java
Gerbangkertosusila	Surabaya City, Sidoarjo Regency, Gresik Regency, Mojokerto Regency, Lamongan Regency, Bangkalan Regency, Mojokerto City	East Java
Tunggal Rogo Mandiri	Tulungagung Regency, Trenggalek Regency, Ponorogo Regency, Madiun Regency, Madiun City, Nganjuk Regency, Kediri Regency, Kediri City	East Java
Sarbagita	Denpasar City, Badung Regency, Gianyar Regency, Tabanan Regency	Bali
Banjarbakula	Banjarbaru City, Banjarmasin City, Banjar Regency, Barito Kuala Regency, Tanah Laut Regency	South Kalimantan
Bimindo	Manado City, Bitung City, Tomohon City, Minahasa Regency, North Minahasa Regency	North Sulawesi
Mamminasatapa	Makassar City, Maros Regency, Gowa Regency, Takalar Regency, Pangkajene Islands Regency	South Sulawesi

Indonesia provides a compelling case study because it is one of the fastest urbanising nations in Southeast Asia, with a rapidly growing population and pressing transportation issues, including congestion, air pollution, and high dependence on private vehicles. By 2023, more than 58.57% of the population resided in urban areas, up from 51% in 2010. This figure is projected to reach 59.6% by 2025, equivalent to nearly 171 million urban residents, and approximately 70% by 2045 (around 223 million people) [8, 9]. This demographic transformation intensifies demand for housing, commercial facilities, and transportation infrastructure, with estimates suggesting a need for 13 million new housing units by 2025. Such rapid growth has exacerbated traffic congestion in major cities. For instance, Jakarta ranks among the most congested cities globally, with one in four workers spending at least 90 minutes commuting one way [150]. This congestion contributes to severe air pollution, which is estimated to cause over 10,000 premature deaths annually in Jakarta alone. Moreover, reliance on private vehicles remains dominant, amplifying carbon emissions and undermining sustainability goals. These dynamics make Indonesia not only a critical setting for assessing governance and transport policies but also a strategic example for understanding how urbanisation, technological adoption, and behavioral shifts converge to shape sustainable transport futures in emerging economies.

4.2 Measurement of Variable

The measurement of variables in this study was developed based on the adaptation of indicators from previous studies on sustainable transportation, technology adoption, service quality, and mobility behavior. To ensure

conceptual validity, the initial pool of measurement items was compiled through an extensive review of relevant literature in these domains. Indicators were selected based on their theoretical relevance and empirical use in prior studies. To maintain contextual relevance to the Indonesian metropolitan environment, an expertise Focus Group Discussion (FGD) was conducted involving academics in the field of urban transportation, urban planners, officials from the Transportation Agency, and transportation technology practitioners. During this stage, the experts reviewed the initial list of indicators to evaluate their content validity, contextual suitability, and potential redundancy. Indicators considered overlapping, ambiguous, or less relevant to the Indonesian urban transportation context were removed or refined. This process helped ensure that the selected indicators adequately represented the conceptual constructs while maintaining clarity and relevance for respondents.

In addition, a pilot test was conducted with a small group of respondents to evaluate the clarity and reliability of the questionnaire items. Feedback from the pilot test was used to refine wording and ensure that the indicators were easily understood by respondents. This procedure also helped to minimize redundancy among items and improve the overall measurement quality before the full survey was administered.

The Sustainable Urban Transport Policy variable is measured by 14 indicators representing environmental, social, and economic dimensions [19, 64, 151, 152]. The Technology Adoption in Urban Transport variable consists of 15 indicators, including infrastructure readiness, user acceptance, technology integration, service quality, and contextual factors [22, 23, 57, 65, 153]. The Public Attitude Toward Sustainable Mobility variable has 10 indicators that assess attitudes, awareness, and social values towards environmentally friendly mobility [47, 66, 86, 124, 154]. The variables of Perceived Service Quality of Public Transport are measured by 10 indicators, including convenience, security, timeliness, accessibility, and reliability of services [52, 85, 135, 155, 156]. The Actual Sustainability Public Transport Use Behavior variable includes seven indicators of real behavior in public transportation use [17, 31, 69, 71].

The Urban Density variable was constructed using secondary data obtained from Statistics Indonesia (BPS), regional spatial planning documents (RTRW), and transportation statistics from local transportation agencies (Dishub). The data refer to the most recent datasets available during 2024–2025 and were compiled at the metropolitan area level, which serves as the spatial unit of analysis in this study. Urban Density was measured using four indicators: population density (people per km²), built-up area (% of total urban land), land use intensity (ratio of building units per hectare), and density of public transport infrastructure, measured by the number of stations and major road/rail networks per km² [136–138, 157]. All survey instruments (except Urban Density) used a 5-point Likert scale (1 = strongly disagree to 5 = strongly agree) in Indonesian to measure respondents' perceptions, attitudes, and behaviors consistently.

4.3 Data Sample and Collection Procedures

The study population comprises users of public transportation within Indonesia's officially designated metropolitan areas. A purposive sampling approach was chosen to deliberately recruit respondents with direct, regular experience of urban public transit. Eligibility required participants to be at least 18 years old, live in one of the listed metropolitan regions, and use public transport a minimum of three times per week. Individuals who mainly relied on private vehicles or who used public transport only sporadically were excluded to preserve sample relevance.

Following guideline of using five respondents per measurement indicator, a total of 60 indicators would require a minimum sample size of 300 respondents [158]. Meanwhile, an a priori power analysis conducted using G*Power version 3.1.9.7 ($f^2 = 0.20$, $\alpha = 0.05$, power = 0.92, with seven predictors) resulted in a smaller minimum sample requirement of 140 respondents [159]. To be conservative and increase the robustness of partial least squares-structural equation modelling (PLS-SEM) estimates, the study targets at least 300 valid responses.

Data were collected between July and September 2025 using Google Forms as the primary survey instrument. The dissemination strategy combined multiple channels: targeted research promotion through social media platforms and professional communication networks, complemented by large-scale online distribution supported by a credible independent survey institution. This mixed dissemination ensured broad coverage across metropolitan regions while also maintaining data quality and respondent diversity. All participants were provided with clear information regarding the purpose of the study, and ethical protocols were strictly followed, including informed consent, voluntary participation, and the protection of anonymity and confidentiality.

This study successfully involved 500 respondents with detailed in Table 2, where the majority of respondents were men (64.6%), with the dominance of the productive age group of 25–34 years (42.8%). The education level of most of the respondents is relatively high, with almost half (49.4%) having a bachelor's degree (S1), indicating the participation of educated people in sustainable transportation issues. Most of them work in the private sector (47.6%) with a monthly income between IDR 3–9 million (69%). In terms of mobility, the main modes of transportation used are commuter trains (33.2%) and online-based transportation (31.6%), indicating the adoption of technology in urban travel behavior. The dominant travel destination is to work (59.4%), with a frequency of 5–6 times per week (43.4%). Most live in the Greater Jakarta area (54.4%), which represents the main metropolitan characteristics in Indonesia in sustainable transportation policies.

Table 2. Characteristic of the respondent

Variable	Category	Frequency (<i>n</i>)	Percentage (%)
Gender	Male	323	64.6
	Female	177	35.4
Age group (years)	18–24	63	12.6
	25–34	214	42.8
	35–44	146	29.2
	45–54	54	10.8
	≥55	23	4.6
	High school or below	71	14.2
Educational level	Diploma (D1–D3)	93	18.6
	Bachelor's degree (S1)	247	49.4
	Master's degree (S2)	76	15.2
	Doctoral degree (S3)	13	2.6
Employment status	Government employee	73	14.6
	Private sector employee	238	47.6
	Entrepreneur/self-employed	97	19.4
	Student	56	11.2
Monthly income level (in IDR)	Other	36	7.2
	<3,000,000	58	11.6
	3,000,000–5,999,999	169	33.8
	6,000,000–8,999,999	176	35.2
	≥9,000,000	97	19.4
Primary mode of public transport used	Bus (BRT/regular)	92	18.4
	Commuter train (KRL/MRT/LRT)	166	33.2
	Angkot/minibus	59	11.8
	Online-based transport (Gojek/Grab)	158	31.6
	Others	25	5.0
Trip purpose (dominant)	Commuting to work	297	59.4
	School/university	53	10.6
	Leisure/recreation	67	13.4
	Errands/shopping	61	12.2
	Other	22	4.4
Trip frequency (times per week)	3–4	118	23.6
	5–6	217	43.4
	≥7	165	33.0
Average one-way travel time	<30 minutes	93	18.6
	30–59 minutes	206	41.2
	60–89 minutes	134	26.8
	≥90 minutes	67	13.4
Metropolitan area of residence	Jabodetabekpunjur	272	54.4
	Mebidangro	41	8.2
	Patungrava Agung	26	5.2
	Cekungan Bandung	39	7.8
	Kedungsepur	27	5.4
	Gerbangkertosusila	43	8.6
	Tunggal Rego Mandiri	11	2.2
	Sarbagita	15	3.0
	Banjarbakula	9	1.8
	Bimindo	10	2.0
	Mamminasatapa	7	1.4

Note: IDR = Indonesian Rupiah; BRT = Bus Rapid Transit; KRL = Commuter Line Electric Train (Kereta Rel Listrik); MRT = Mass Rapid Transit; LRT = Light Rail Transit.

4.4 Data Analysis Techniques

The study applies PLS-SEM with the aid of SmartPLS 4 software. This technique was chosen because it is highly appropriate for predictive, exploratory research that involves multiple latent constructs, complex relationships, and

data that may not meet the assumption of normality [160]. PLS-SEM is also advantageous when the theoretical framework is still under development and when the focus extends to both explanation and prediction of behavioral outcomes [161]. Compared with covariance-based SEM (CB-SEM), PLS-SEM is more suitable for models with many indicators and hierarchical relationships, and it places greater emphasis on maximising the explained variance of endogenous constructs rather than reproducing the covariance matrix [162]. In addition, PLS-SEM performs well in studies that aim to identify key drivers of behavior and generate managerial insights, making it appropriate for analysing sustainable transportation behavior in complex metropolitan contexts.

The analytical procedure is carried out in three main stages. First, the measurement model (outer model) is validated by testing indicator reliability, convergent validity, and discriminant validity [163]. Second, the structural model (inner model) is evaluated through path coefficients (β), explained variance (R^2), effect sizes (f^2), and predictive relevance (Q^2) [162]. Finally, several advanced procedures are employed: moderation analysis for conditional effects, importance-performance mapping analysis (IPMA) for managerial prioritisation [164], partial least squares predict (PLSpredict) to examine predictive capability [165], and finite mixture partial least squares (FIMIX-PLS) to detect hidden heterogeneity within the sample [166]. This stepwise approach strengthens both the methodological rigour and the practical insights of the study.

5 Results

5.1 Common Methods Bias

To address potential common method bias (CMB), this study applied the full collinearity assessment approach proposed by Kock [167]. This method examines the variance inflation factor (VIF) values for all latent constructs simultaneously. If the VIF values are below the recommended threshold of 5, both multicollinearity and CMB are considered unlikely to pose a serious threat to the validity of the model. As shown in Table 3, all constructs exhibit VIF values below 5. In addition, a random marker variable was included in the analysis to further assess potential method bias. The results indicate that CMB is not a significant concern in this study.

Table 3. Full collinearity assessment for common method bias (CMB)

Construct	Full Collinearity Variance Inflation Factor (VIF)
Sustainable Urban Transport Policy	4.451
Technology Adoption in Urban Transport	4.763
Public Attitude Toward Sustainable Mobility	3.752
Perceived Service Quality of Public Transport	4.229
Actual Sustainable Public Transport Use Behavior	3.901
Urban Density	2.592
Random Marker Variable	1.874

5.2 Measurement Outer Model

The outer model measurement stage in the PLS-SEM analysis is used to assess the extent to which the indicator can reflect the latent construct being measured [160]. This process ensures that each indicator has consistency and accuracy in describing the conceptual variables before entering the structural model stage. The test was carried out through convergent validity, discriminant validity, construct reliability analysis (through Composite Reliability and Cronbach's alpha), and multicollinearity checks with VIF values.

The results in Table 4 show that all indicators have a loading factor value above 0.70 and an average variance extracted (AVE) greater than 0.50 [160], which signifies strong convergent validity. Cronbach's alpha values range from 0.880–0.949 and Composite Reliability between 0.907–0.974, confirming that the construct is reliable above 0.7 [168]. In addition, all VIF values below 5 indicate the absence of symptoms of multicollinearity [169]. Thus, the measurement model has met the requirements for validity and reliability, and is suitable to proceed to the structural analysis stage of the model in PLS-SEM.

Table 4. Full collinearity assessment for common method bias (CMB)

Variable	Factor Loadings	Average Variance Extracted (AVE)	Cronbach's Alpha	Composite Reliability	Variance Inflation Factor (VIF)
Sustainable Urban Transport Policy (SUTP)					
SUTP1	0.721				4.189
SUTP2	0.862				4.453
SUTP3	0.916				3.795
SUTP4	0.996				3.604
SUTP5	0.951				3.528
SUTP6	0.788				4.619
SUTP7	0.713	0.627	0.927	0.974	4.036
SUTP8	0.802				3.731
SUTP9	0.963				3.714
SUTP10	0.737				3.280
SUTP11	0.809				2.931
SUTP12	0.888				3.822
SUTP13	0.813				4.544
SUTP14	0.839				4.425
Technology Adoption in Urban Transport (TAUT)					
TAUT1	0.719				3.953
TAUT2	0.702				4.557
TAUT3	0.829				3.618
TAUT4	0.960				4.382
TAUT5	0.894				3.915
TAUT6	0.722				4.275
TAUT7	0.817				3.712
TAUT8	0.835	0.563	0.880	0.907	3.115
TAUT9	0.920				3.945
TAUT10	0.900				4.733
TAUT11	0.916				4.177
TAUT12	0.887				3.811
TAUT13	0.868				4.017
TAUT14	0.886				4.760
TAUT15	0.770				4.727
Public Attitude Toward Sustainable Mobility (PASM)					
PASM1	0.877				3.392
PASM2	0.868				3.750
PASM3	0.734				3.508
PASM4	0.825				3.429
PASM5	0.920	0.646	0.949	0.956	3.156
PASM6	0.789				3.189
PASM7	0.805				3.531
PASM8	0.928				3.394
PASM9	0.863				2.832
PASM10	0.956				2.613
Perceived Service Quality of Public Transport (PSQ)					
PSQ1	0.866				3.135
PSQ2	0.939				3.255
PSQ3	0.781				3.744
PSQ4	0.897				3.560
PSQ5	0.830	0.670	0.912	0.936	3.696
PSQ6	0.753				3.556
PSQ7	0.755				4.222
PSQ8	0.811				3.748
PSQ9	0.907				4.218
PSQ10	0.811				3.255

Variable	Factor Loadings	Average Variance Extracted (AVE)	Cronbach's Alpha	Composite Reliability	Variance Inflation Factor (VIF)
Actual Sustainability Public Transport Use Behavior (ASPT)					
ASPT1	0.806				3.737
ASPT2	0.745				3.904
ASPT3	0.899				3.261
ASPT4	0.851	0.641	0.945	0.954	3.859
ASPT5	0.761				2.518
ASPT6	0.846				2.866
ASPT7	0.802				2.791
Urban Density					
Population density	0.970				2.532
Built-up area	0.984	0.741	0.884	0.921	1.957
Land use intensity	0.824				2.451
Transportation infrastructure density	0.915				2.595

The discriminant validity stage in PLS-SEM is used to ensure that each latent construct is genuinely unique and distinguishable from other constructs [160]. This validity assesses the extent to which a variable only measures the concept in question, not other constructs that have conceptual similarities. Based on Table 5, the results of the Fornell-Larcker Criterion test show that the square root of AVE of each construct is greater than the correlation between other constructs, which indicates an adequate separation of concepts [170]. The value of the Heterotrait-Monotrait Ratio (HTMT) is entirely below the limit of 0.90 [171]. It asserts that the relationship between constructs does not show conceptual overlap. Thus, the model has good discriminant validity, indicating that each variable has a clear identity and is distinguishable from the others.

Table 5. Outer model analysis: discriminant validity

	Fornell-Larcker Criterion					
	ASPT	PSQ	PASM	SUTP	TAUT	UD
ASPT	0.964					
PSQ	0.850	0.872				
PASM	0.841	0.748	0.928			
SUTP	0.846	0.757	0.844	0.894		
TAUT	0.741	0.757	0.834	0.771	0.873	
UD	0.796	0.760	0.783	0.871	0.847	0.861
Heterotrait-Monotrait Ratio (HTMT)						
ASPT						
PSQ	0.795					
PASM	0.791	0.784				
SUTP	0.687	0.786	0.674			
TAUT	0.778	0.784	0.761	0.795		
UD	0.675	0.820	0.762	0.828	0.798	

Note: ASPT = Actual Sustainability Public Transport Use Behavior; PSQ = Perceived Service Quality of Public Transport; PASM = Public Attitude Toward Sustainable Mobility; SUTP = Sustainable Urban Transport Policy; TAUT = Technology Adoption in Urban Transport; UD = Urban Density (measured by population density, built-up area, land use intensity, and transportation infrastructure density).

5.3 Inner Model and Structural Model

The Inner Model or Structural Model stage in PLS-SEM aims to test the relationships between latent constructs that have been stated in the research hypothesis. This evaluation involves testing the strength and significance of the causal pathway using a bootstrapping procedure (usually 5,000 resampling) to obtain the β , t -value, and p -value

values [172]. These values indicate the direction, intensity, and level of significance of the relationship between variables.

The results of direct hypothesis testing in Table 6 show that most of the relationships between constructs are significant at a 95% confidence level. Sustainable Urban Transport Policy has a positive and significant effect on Public Attitude Toward Sustainable Mobility ($\beta = 0.622$; $p < 0.001$) and Perceived Service Quality ($\beta = 0.485$; $p < 0.001$). Technology Adoption also had a positive effect on Public Attitude ($\beta = 0.300$; $p < 0.001$) and Service Quality ($\beta = 0.470$; $p < 0.001$). However, the impact of Public Attitude on Actual Sustainable Transport Behavior was not significant ($\beta = 0.347$; $p = 0.078$), while Perceived Service Quality showed a significant positive effect on such behavior ($\beta = 0.442$; $p = 0.023$). These results indicate that service quality has a stronger role in encouraging sustainable transportation behavior than public attitudes alone.

Table 6. Inner analysis: results of the direct hypothesis

No.	Hypothesis Statement	Path Coefficient (β)	t-Value	p-Value	Confidence Interval (2.5%–97.5%)	Decision	f^2 Effect Size
H1	Sustainable Urban Transport Policy → Public Attitude Toward Sustainable Mobility	$\beta_1 = 0.622$	7.371	0.000	0.443–0.775	Supported	0.121
H2	Sustainable Urban Transport Policy → Perceived Service Quality of Public Transport	$\beta_2 = 0.485$	6.043	0.000	0.317–0.633	Supported	0.119
H3	Technology Adoption in Urban Transport → Public Attitude Toward Sustainable Mobility	$\beta_3 = 0.300$	3.593	0.000	0.148–0.476	Supported	0.188
H4	Technology Adoption in Urban Transport → Perceived Service Quality of Public Transport	$\beta_4 = 0.470$	6.022	0.000	0.326–0.631	Supported	0.211
H5	Public Attitude Toward Sustainable Mobility → Actual Sustainable Public Transport Use Behavior	$\beta_5 = 0.347$	1.764	0.078	-0.092–0.680	Not Supported	0.053
H6	Perceived Service Quality of Public Transport → Actual Sustainable Public Transport Use Behavior	$\beta_6 = 0.442$	2.275	0.023	0.117–0.879	Supported	0.079

The R^2 and Q^2 test stages were used to evaluate the predictive capabilities of structural models in PLS-SEM. The R^2 value indicates the proportion of endogenous construct variance that the exogenous construct can explain, while Q^2 assesses the predictive relevance of the model using the blindfolding technique [173]. The higher the value of these two indicators, the better the model's ability to explain and predict the variables being studied.

The results in Table 7 show a very high R^2 value, namely 0.896 for Public Attitude Toward Sustainable Mobility, 0.928 for Perceived Service Quality, and 0.932 for Actual Sustainable Transport Behavior. These values indicate that the model has powerful explanatory abilities (>0.75). In addition, the high Q^2 value (0.761–0.792) confirms that the model has substantial predictive relevance. Overall, structural models show excellent performance in predicting

and explaining sustainable public transportation behavior in Indonesia's metropolitan areas.

Table 7. R^2 and Q^2 testing result

Variable Endogenous	R^2	Adjusted R^2	Q^2
Public Attitude Toward Sustainable Mobility	0.896	0.896	0.761
Perceived Service Quality of Public Transport	0.928	0.928	0.789
Actual Sustainable Public Transport Use Behavior	0.932	0.931	0.792

5.4 Moderation Analysis

The study also examined the role of Urban Density as a moderating variable that can strengthen or weaken the relationship between psychological factors and sustainable public transportation use behavior. This moderation analysis aims to understand how Urban Density affects the dynamics of adopting environmentally friendly transportation behavior in Indonesia's metropolitan areas.

Based on Table 8, the results show that Urban Density significantly moderates the relationship between Public Attitude Toward Sustainable Mobility and Actual Sustainable Transport Behavior ($\beta = 0.122$; $t = 2.412$; $p = 0.012$), thus the H7 hypothesis is supported. This indicates that the influence of positive public attitudes on sustainable transportation behavior is stronger in high-density areas. In contrast, the effect of Urban Density moderation on the relationship between Perceived Service Quality and Actual Sustainable Behavior was not significant ($\beta = 0.013$; $p = 0.733$), suggesting that the quality of transportation services was not significantly affected by variations in regional density.

Table 8. Moderation testing results

No.	Hypothesis Statement	Path Coefficient (β)	t -Value	p -Value	Confidence Interval (2.5%–97.5%)	Decision	f^2 Effect Size
H7	Urban Density $\times$ Public Attitude Toward Sustainable Mobility $\rightarrow$ Actual Sustainable Public Transport Use Behavior	0.122	2.412	0.012	0.076–0.263	Supported	0.093
H8	Urban Density $\times$ Perceived Service Quality of Public Transport $\rightarrow$ Actual Sustainable Public Transport Use Behavior	0.013	0.341	0.733	-0.083–0.063	Not Supported	0.004

5.5 Importance-Performance Mapping Analysis

IPMA is used to identify strategic priorities in improving target variables [164], namely Actual Sustainable Public Transport Use Behavior. This analysis combines the importance and performance levels to determine which constructs are most influential and need attention in sustainable transportation policies and implementations.

Based on Figure 2 and Table 9, the IPMA results show that Perceived Service Quality (importance = 0.442; performance = 64,331) has the most significant influence on sustainable public transportation use behavior, followed by Sustainable Urban Transport Policy (0.430; 64,407). Both are the main factors that need to be improved to strengthen the sustainable behavior of the community. Public Attitude Toward Sustainable Mobility (0.347; 63,546) and Technology Adoption (0.312; 64,781) also made significant contributions, but with relatively stable performance. Meanwhile, Urban Density showed the lowest influence value (0.284; 62.263), indicating that urban spatial factors

have an effect not as significant as the aspects of policies, services, and technology adoption in encouraging sustainable transportation behaviors.

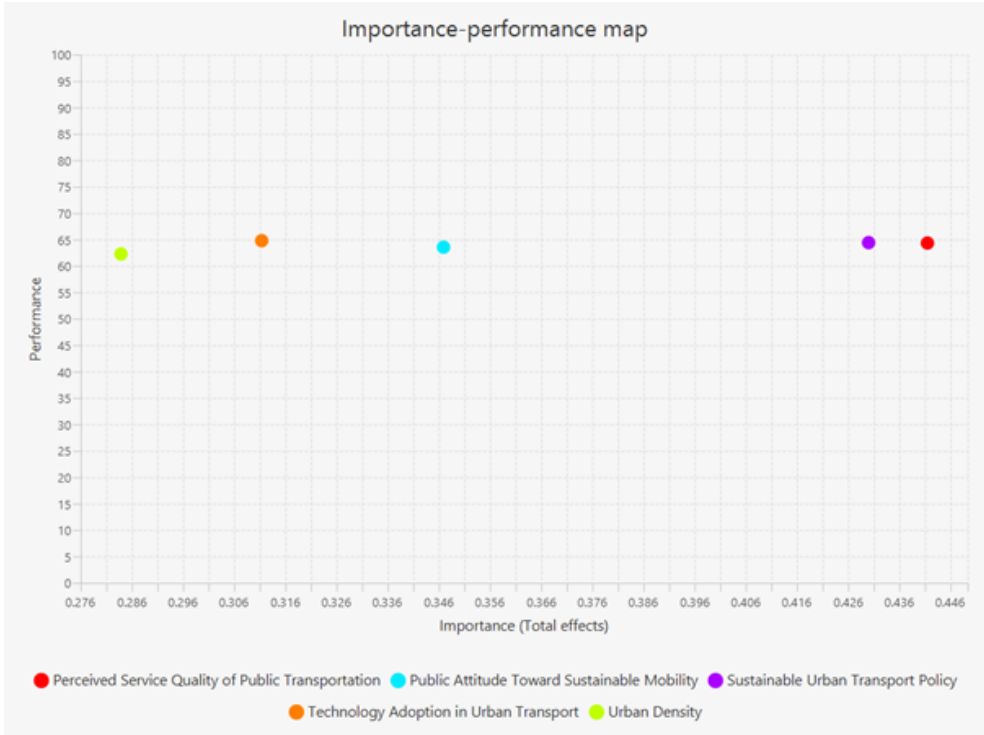


Figure 2. Importance-performance mapping analysis (IPMA) matrix

Table 9. Importance-performance mapping analysis (IPMA) testing results (construct target: actual sustainable public transport use behavior)

	Importance	Performance
Perceived Service Quality of Public Transportation	0.442	64.331
Public Attitude Toward Sustainable Mobility	0.347	63.546
Sustainable Urban Transport Policy	0.430	64.407
Technology Adoption in Urban Transport	0.312	64.781
Urban Density	0.284	62.263

5.6 Finite Mixture Analysis

FIMIX-PLS is an advanced approach in PLS-SEM used to identify the presence of non-observable heterogeneity in data [174]. Theoretically, FIMIX allows researchers to detect segments or groups of respondents that have different structural relationship patterns despite coming from the same population [175]. This method helps uncover hidden variations in behaviors, perceptions, or preferences that traditional PLS models can't explain. The results in Table 10 show two segments, namely Segment 1 by 83% and Segment 2 by 17%. This proportion signifies that most respondents have similar behavioral characteristics, while other small groups show different patterns that can be further analysed.

Table 10. Segment size

Segment 1	Segment 2
0.830	0.170

Furthermore, the results of the FIMIX β in Table 11 show that there is a difference in the pattern of relationships between constructs in the two respondent segments. The results suggest that Segment 1, which represents the majority of respondents (83%), is largely composed of individuals from highly urbanised metropolitan regions—particularly

the Greater Jakarta area—with frequent commuting patterns and relatively stable employment in the private or government sector. These respondents tend to rely on structured public transport systems such as commuter trains, MRT, or BRT, which may explain their stronger responsiveness to sustainable transport policies and positive attitudes toward sustainable mobility.

Table 11. Path coefficient (β)-finite mixture (FIMIX) segment

	Segment 1	Segment 2
Sustainable Urban Transport Policy → Perceived Service Quality of Public Transportation	0.540	0.242
Sustainable Urban Transport Policy → Public Attitude Toward Sustainable Mobility	0.541	0.455
Technology Adoption in Urban Transport → Perceived Service Quality of Public Transportation	0.446	0.584
Technology Adoption in Urban Transport → Public Attitude Toward Sustainable Mobility	0.444	0.220
Public Attitude Toward Sustainable Mobility → Actual Sustainability Public Transport Use Behavior	0.512	-0.252
Perceived Service Quality of Public Transportation → Actual Sustainability Public Transport Use Behavior	0.305	0.995
Urban Density × Public Attitude Toward Sustainable Mobility → Actual Sustainability Public Transport Use Behavior	0.017	0.066
Urban Density × Perceived Service Quality of Public Transportation → Actual Sustainability Public Transport Use Behavior	0.017	-0.094

In contrast, Segment 2 (17%) appears to include respondents whose travel behavior is more strongly influenced by technological convenience rather than policy or attitudinal factors. This group shows relatively higher reliance on app-based mobility services and flexible travel arrangements, which may explain why technology adoption has a stronger influence on perceived service quality in this segment. The weaker relationship between public attitude and actual behavior in this group suggests that practical considerations, such as convenience and technological accessibility, may play a more dominant role than normative or policy-driven motivations.

These findings provide a more nuanced understanding of urban mobility behavior, indicating that while most users respond positively to sustainable transport policies and service improvements, a smaller group of users is more strongly influenced by technological innovation and service convenience. This segmentation highlights the importance of designing differentiated policy and technological strategies to accommodate diverse user preferences within urban transportation systems.

The results of the R^2 FIMIX in Table 12 show that there is a variation in the explanatory strength of the model between segments. In Segment 1, the R^2 value was very high throughout the construct, namely 0.972 for Public Attitude, 0.970 for Perceived Service Quality, and 0.961 for Actual Sustainable Transport Behavior. This indicates that the model can explain more than 95% of the variation in sustainable transport behavior in this group, demonstrating high consistency between policy, technology adoption, and public response to sustainable transport. In contrast, Segment 2 had much lower R^2 values, 0.622, 0.771, and 0.833, respectively, suggesting that the influence of exogenous constructs on endogenous variables was weaker. This difference confirms the existence of structural heterogeneity among the respondent groups. Thus, Segment 1 describes a group with a strong and consistent explanatory model, while Segment 2 shows the instability of the relationship between variables.

Table 12. R^2 -finite mixture (FIMIX) segment

	Original Sample R^2	Weighted Average R^2	Segment 1	Segment 2
Actual Sustainability Public Transport Use Behavior	0.932	0.940	0.961	0.833
Perceived Service Quality of Public Transportation	0.928	0.936	0.970	0.771
Public Attitude Toward Sustainable Mobility	0.896	0.912	0.972	0.622

5.7 Partial Least Squares Predict

PLSPredict is used in PLS-SEM to assess the model's predictive ability against new data. Theoretically, this method compares the performance of the PLS-SEM model with the benchmark model (linear model, i.e., LM) through metrics such as root mean square error (RMSE) and mean absolute error (MAE) [165]. The results in Table 13 show that most indicators have a positive Q^2 predict value (0.379–0.741), indicating good predictive ability. In addition, the PLS-SEM RMSE and MAE values are generally lower than those of LM, indicating that the PLS-SEM model has superior and stable predictive performance in projecting sustainable transportation behavior.

Table 13. Partial least squares predict (PLSPredict) testing results

Variabel	Q^2 Predict	PLS-SEM_RMSE	PLS-SEM_MAE	LM_RMSE	LM_MAE
ASPT1	0.738	0.655	0.565	0.719	0.585
ASPT2	0.735	0.677	0.582	0.727	0.596
ASPT3	0.741	0.653	0.560	0.684	0.566
ASPT4	0.722	0.709	0.600	0.776	0.620
ASPT6	0.579	0.790	0.619	0.910	0.661
ASPT7	0.648	0.744	0.636	0.833	0.654
ASPT8	0.643	0.744	0.604	0.808	0.627
PSQ1	0.635	0.787	0.620	0.793	0.625
PSQ10	0.715	0.665	0.565	0.703	0.582
PSQ2	0.675	0.745	0.605	0.791	0.623
PSQ3	0.709	0.704	0.593	0.790	0.607
PSQ4	0.690	0.756	0.604	0.842	0.652
PSQ6	0.689	0.729	0.595	0.724	0.618
PSQ7	0.704	0.741	0.586	0.812	0.579
PSQ8	0.731	0.686	0.585	0.726	0.607
PSQ9	0.739	0.656	0.559	0.701	0.580
PASM1	0.731	0.661	0.560	0.695	0.559
PASM10	0.495	0.876	0.695	0.946	0.676
PASM2	0.724	0.706	0.584	0.793	0.618
PASM3	0.729	0.708	0.595	0.754	0.599
PASM4	0.719	0.703	0.583	0.735	0.593
PASM5	0.704	0.719	0.603	0.773	0.619
PASM6	0.379	0.886	0.719	1.018	0.673
PASM7	0.441	0.912	0.732	1.023	0.748
PASM8	0.678	0.760	0.616	0.808	0.649

Note: PLS = partial least squares; SEM = structural equation modelling; RMSE = root mean squared error; MAE = mean absolute error; LM = linear model; ASPT = actual sustainability public transport use behavior; PSQ = perceived service quality of public transport; PASM = Public Attitude Toward Sustainable Mobility; SUTP = Sustainable Urban Transport Policy; TAUT = Technology Adoption in Urban Transport; UD = Urban Density (measured by population density, built-up area, land use intensity, and transportation infrastructure density).

6 Discussion and Implications

6.1 Discussion

The results indicate that Sustainable Urban Transport Policy exerts a significant and substantive influence on both public attitudes toward sustainable mobility and perceived service quality of public transport. Prior studies have shown that well-designed policy interventions, service quality improvements, and psychosocial factors jointly shape citizens' pro-sustainability orientations and behavioral intentions [63, 75, 86, 87, 176, 177]. These findings suggest that comprehensively designed policies—encompassing emission regulations, modal integration, and incentives for

green transport use—not only establish a structural framework but also foster cognitive and affective shifts in public behavior toward environmentally responsible mobility. The strong β in H1 highlights the transformative capacity of policy dimensions in shaping new social norms and collective awareness of sustainability, aligning with the behavioral governance approach in contemporary urban studies.

Meanwhile, the significant effect of policy on perceived service quality of public transport (H2) demonstrates that coherent and measurable policy design contributes to improved user experience. Empirical evidence supports this relationship, indicating that sustainable transport policies significantly shape users' perceptions of service quality through improvements in reliability, comfort, safety, and accessibility [85, 95, 96, 98, 151, 178]. Policies that enhance connectivity, comfort, and system reliability generate positive perceptions of service quality, which in turn strengthen the intention to use public transport. This effect reflects the reciprocal relationship between policy coherence and service perception, wherein sustainability is determined not only by physical infrastructure but also by perceptions of efficiency and equity in public service delivery. These results affirm that the success of the transition toward sustainable mobility in Indonesia's metropolitan areas critically depends on the integration between visionary macro-level policies and micro-level strategies that shape public perception and attitude.

The findings reveal that Technology Adoption in Urban Transport plays a pivotal role in shaping both public attitudes toward sustainable mobility and perceived service quality of public transport. The significant yet moderate influence of technology adoption on public attitudes (H3) highlights that technological interventions such as real-time travel information, mobile ticketing, and digital navigation systems contribute to enhancing awareness, trust, and perceived convenience among citizens. By improving transparency and user control, technology fosters positive affective and cognitive evaluations toward sustainable travel options. This aligns with contemporary models of technology-enabled behavioral change, where digital integration acts as a catalyst for shifting mobility preferences toward more sustainable practices. Empirical studies corroborate this finding, showing that technology adoption significantly shapes public attitudes toward sustainable transport through perceived usefulness, ease of use, and trust [43, 104–106, 179].

The strong relationship between technology adoption and perceived service quality (H4) underscores the strategic importance of digital transformation in elevating public transport performance. Technological innovations not only streamline operations and reduce uncertainty but also enhance perceived efficiency, safety, and reliability—key dimensions of user satisfaction in the public transport experience. This finding supports the argument that service digitalisation is integral to the sustainability transition, bridging the gap between system performance and user perception. These outcomes emphasise that the integration of technology in urban mobility governance is not merely an operational upgrade but a behavioral mechanism that reinforces sustainable mobility systems. It demonstrates how innovative mobility ecosystems can enhance both service quality and pro-environmental public attitudes. The findings enrich the discourse on digital urban governance in developing contexts, highlighting technology as an enabler of participatory, adaptive, and sustainability-oriented behavioral transformation. Empirical evidence substantiates this finding, indicating that technology adoption significantly enhances perceived service quality through improvements in punctuality, reliability, comfort, and safety [98, 115, 116]. Real-time tracking and automated scheduling notably reduce uncertainty and elevate user satisfaction across diverse urban contexts.

The results indicate that Public Attitude Toward Sustainable Mobility does not significantly influence actual sustainable public transport use behavior. Contrary to this result, prior studies have generally found a significant link between public attitudes and sustainable mobility behavior, where positive attitudes reduce private car dependence and enhance public transport use [123–127]. This finding highlights a critical attitude–behavior gap, a well-documented phenomenon in sustainability research, where positive attitudes toward environmentally friendly practices do not necessarily translate into consistent behavioral outcomes. In the Indonesian metropolitan context, this gap may stem from structural and contextual constraints—such as limited service coverage, inadequate reliability, and social preferences for private vehicles—that inhibit behavioral realisation despite favourable attitudes. The result suggests that fostering sustainable mobility behavior requires more than cultivating awareness or positive sentiment; it demands systemic reinforcement through institutional, infrastructural, and normative supports. While individuals may value sustainability, their behavioral choices remain bounded by perceived convenience, safety, and accessibility of alternatives. This underscores the importance of embedding behavioral interventions—such as incentives, behavioral nudges, and improved service experiences—within governance frameworks to convert pro-sustainability attitudes into tangible actions.

The results confirm that perceived service quality of public transport significantly influences actual sustainable public transport use behavior, underscoring the centrality of user experience in driving behavioral change toward sustainable mobility. Empirical evidence supports this result, showing that perceived service quality—encompassing reliability, comfort, safety, and accessibility—strongly predicts satisfaction and actual use of public transport [52, 86, 94, 129]. This finding highlights that individuals' decisions to adopt public transport are not merely normative or attitudinal but are shaped mainly by experiential and performance-based evaluations—including reliability, comfort, safety, and accessibility. When service quality meets or exceeds expectations, it strengthens user trust and habitual

reliance on public transport, fostering a behavioral shift away from private vehicle dependence. This result aligns with the service-dominant logic in urban mobility research, emphasising that perceived value creation through superior service quality is essential to sustain behavioral transformation. In the Indonesian metropolitan context, where infrastructural and operational inconsistencies persist, enhancing perceived quality becomes a strategic entry point for promoting long-term modal shifts. Furthermore, the finding illustrates that behavioral change is contingent upon the alignment between system performance and user perception—indicating that policy interventions must prioritise user-centred service improvements as a behavioral lever. Ultimately, this relationship reinforces that sustainable mobility governance should focus not only on policy design or attitudinal change but on tangible service quality enhancements capable of converting sustainability ideals into everyday transport practices.

This study proposed Urban Density as a moderating factor in the relationship between attitudinal and perceptual antecedents and actual sustainable public transport use behavior, reflecting the spatial dimension of behavioral transformation in metropolitan contexts. The supported moderation effect in H7 indicates that higher Urban Density strengthens the influence of Public Attitude Toward Sustainable Mobility on actual transport behavior. This suggests that in dense urban environments—where accessibility, proximity, and transport interconnectivity are greater—positive attitudes are more easily translated into sustainable behavioral practices. In such contexts, the availability of frequent transit services, shorter travel distances, and better multimodal connectivity reduces the behavioral barriers between intention and action. Dense spatial configurations therefore create structural opportunities that enable individuals to act on their pro-sustainability intentions, reinforcing the role of the built environment as a behavioral facilitator rather than a neutral backdrop.

Conversely, the non-significant moderation effect in H8 implies that Urban Density does not significantly alter the relationship between perceived service quality and transport use behavior. One possible explanation is that perceived service quality reflects operational aspects of transport systems—such as reliability, safety, punctuality, and comfort—which are experienced similarly by users regardless of spatial density. In other words, even in lower-density metropolitan areas, improvements in service performance may directly influence usage behavior without relying on spatial conditions. This asymmetry indicates that attitude-driven behavior tends to be context-sensitive and spatially enabled, whereas service-quality evaluations operate more universally across urban environments. These findings highlight that urban form interacts differently with psychological and experiential drivers, emphasising the importance of integrating spatial planning with behavioral and service-oriented transport policies to support sustainable mobility transitions.

6.2 Theoretical Implications

This study advances urban transport and behavioral governance theory by empirically integrating policy effectiveness and technology adoption frameworks within the TPB. The results empirically validate that sustainable urban transport policies and technology adoption act as dual institutional and experiential levers shaping both cognitive (attitude) and affective (service quality perception) dimensions of commuter behavior. By demonstrating that perceived service quality exerts a more substantial influence on actual behavioral outcomes than attitudes, this study advances the argument that behavioral transformation in mobility is primarily performance-driven rather than purely normative. This challenges conventional models emphasising attitudinal determinants, positioning service experience as a central mediator linking governance design and behavioral realisation.

Furthermore, the identification of an attitude–behavior gap in Indonesia’s metropolitan settings extends the TPB by emphasising the moderating role of structural and contextual barriers, such as infrastructure and socio-normative constraints. The moderating effect of Urban Density highlights spatial context as an enabling condition that translates pro-sustainability cognition into tangible action, thereby enriching urban behavioral theories with spatial-behavioral interdependencies. Collectively, these findings reinforce a multi-level theoretical synthesis where governance quality, technological integration, and urban form jointly determine the translation of sustainability intentions into everyday mobility practices—advancing a context-sensitive model of sustainable behavior in emerging urban systems.

6.3 Practical and Policy Implications

The findings of this study provide critical insights for policymakers, urban planners, and metropolitan authorities in designing evidence-based strategies for sustainable mobility transitions in rapidly urbanising regions. First, the strong influence of Sustainable Urban Transport Policy on both public attitudes and perceived service quality underscores the importance of consistent policy implementation—beyond regulation—to include service improvement, multimodal integration, and inclusive governance. Metropolitan governments must prioritise investment in high-quality, reliable, and affordable public transport systems supported by precise institutional coordination across municipalities.

Second, the significant role of technology adoption in shaping both perceptions and satisfaction suggests that digital innovation—such as e-ticketing, real-time tracking, and MaaS platforms—should be mainstreamed into transport policy frameworks. These tools enhance transparency, efficiency, and trust, thereby translating public

intention into actual behavioral change. Collaboration between public and private sectors becomes essential to ensure scalability, cybersecurity, and user inclusivity, particularly for vulnerable and low-income commuters.

Third, the empirical evidence that perceived service quality—rather than public attitude alone—directly predicts sustainable transport behavior signals a practical shift in policy focus from awareness campaigns to service excellence. Enhancing comfort, safety, punctuality, and accessibility will yield stronger behavioral responses than normative appeals.

The moderating role of Urban Density emphasises that policy and technology interventions must be spatially adaptive. High-density urban cores require demand management and congestion pricing, while lower-density peripheries need TOD and last-mile connectivity. Collectively, these implications call for metropolitan-level governance reform that integrates digital innovation, service quality enhancement, and adaptive spatial planning. Such an integrated governance–technology–behavior framework enables cities to move beyond fragmented interventions toward coherent, citizen-centred sustainable mobility systems aligned with SDG 11 and 13 objectives.

In addition, the IPMA provides more specific guidance for policy prioritisation. The results indicate that Perceived Service Quality demonstrates the highest importance in influencing sustainable public transport use behavior, while its performance remains moderate. This suggests that transport authorities should prioritise practical improvements in service delivery, such as increasing service reliability and punctuality, improving station safety and cleanliness, and ensuring better passenger information systems. Sustainable Urban Transport Policy also shows high importance, highlighting the need for consistent regulatory support for multimodal integration, fare integration across transport modes, and long-term investment in mass transit infrastructure. Meanwhile, Technology Adoption shows relatively stable performance, suggesting that existing digital services—such as mobile ticketing, route planning applications, and real-time vehicle tracking—should be further integrated with public transport operations to enhance user convenience and transparency. These IPMA results provide clearer operational priorities for metropolitan transport authorities in promoting sustainable mobility behavior.

7 Conclusion

This study concludes that the integration between Sustainable Urban Transport Policy and technology adoption plays a vital role in shaping sustainable public transportation behavior in Indonesia's metropolitan areas. The results of the PLS-SEM analysis show that these two factors significantly improve public attitudes and perceived service quality. However, only perceived service quality has a direct effect on actual public transportation use behavior. This finding suggests that changes in mobility behavior are more strongly driven by real service experiences than by positive attitudes alone. One possible explanation is that favourable attitudes toward sustainable mobility do not necessarily translate into actual behavioral change when structural constraints such as accessibility, service coverage, or travel time remain significant barriers. In addition, Urban Density has been shown to moderate the relationship between variables, indicating that the effectiveness of policies and technologies depends on the spatial context and infrastructure of the region. Overall, this study emphasises the importance of collaborative and evidence-based metropolitan governance to optimise policy–technology synergy in driving a behavioral shift toward an inclusive, efficient, and low-emission public transportation system in Indonesia.

This research has several limitations that need to be considered. First, the cross-sectional survey design limits the ability to observe temporal dynamics and causal relationships between variables over time. Longitudinal studies are therefore needed to better capture changes in public transportation behavior alongside policy implementation and technological adoption. Second, although respondents were drawn from several metropolitan areas in Indonesia, more than half are from the Greater Jakarta region, with the dominant age group being 25–34 years. While this reflects the concentration of public transport use in major urban centres, it may limit the representativeness of all metropolitan contexts in Indonesia. Thus, the findings should primarily be interpreted within the context of large metropolitan regions with relatively advanced transport systems. Third, the behavioral variables are based on self-reported perceptions, which may introduce bias. Future research should complement these measures with observational data or intelligent transportation system sensors to strengthen empirical validity. Future studies are encouraged to incorporate multi-level governance perspectives and spatial analysis to better understand how interactions among policy actors, technological innovation, and urban contexts shape sustainable mobility behaviors.

Author Contributions

Conceptualization, S.M. and T.R.; methodology, S.M. and K.C.; software, M.A.; validation, S.M., T.R., and K.C.; formal analysis, S.M. and M.A.; investigation, S.M.; resources, T.R. and K.C.; data curation, M.A.; writing—original draft preparation, S.M.; writing—review and editing, T.R., K.C., and M.A.; visualization, M.A.; supervision, T.R.; project administration, S.M.; funding acquisition, S.M. and T.R. All authors have read and agreed to the published version of the manuscript.

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Data Availability

The data used to support the research findings are available from the corresponding author upon request.

Conflicts of Interest

The authors declare no conflicts of interest.

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Appendix

Table A1. Variable measurement

Variable	Indicator Code	Indicator Statement
Sustainable Urban Transport Policy (SUTP)	SUTP1	Transportation in my city contributes to good air quality (e.g., low vehicle emissions).
	SUTP2	Efforts to reduce CO ₂ emissions from transport activities have been well implemented.
	SUTP3	The transport system in my city is energy-efficient.
	SUTP4	Urban traffic noise levels are well controlled.
	SUTP5	The transport system in my city is safe and minimizes traffic accident risks.
	SUTP6	Public transport (bus, train, etc.) is punctual and reliable.
	SUTP7	Public transport services in my city have wide coverage.
	SUTP8	Public transport is accessible to vulnerable groups (e.g., people with disabilities, elderly, and low-income residents).
	SUTP9	Transport policies in my city support equity and fair access for all citizens.
	SUTP10	The cost of using public transport is affordable for the public.
	SUTP11	The transport system supports travel time efficiency (e.g., reduces congestion).
	SUTP12	Transport policies help reduce operational costs for private vehicle users.
	SUTP13	Public transport increases community productivity (e.g., supports work and economic activities).
	SUTP14	Investment in public transport provides economic benefits for the city as a whole.
Technology Adoption in Urban Transport (TAUT)	TAUT1	My city has adequate digital infrastructure (e.g., sensors, IoT networks, AI) to support smart transport.
	TAUT2	The availability of electric vehicle (EV) charging stations supports EV use.
	TAUT3	The city government actively supports smart transport technologies through policy and facilities.
	TAUT4	I find urban transport technologies beneficial for improving my daily mobility.
	TAUT5	New transport technologies (e.g., MaaS apps, eticketing) are easy to use.
	TAUT6	I am willing to use technology-based transport services (e.g., ride-sharing, microtransit, autonomous vehicles) if available.
	TAUT7	My social environment (family/friends/community) encourages the adoption of new transport technologies.
	TAUT8	I have used digital-based transport applications (e.g., e-ticketing, Mobility-as-a-Service).
	TAUT9	I support the integration of public transport with digital platforms (e.g., one app for bus, train, and ride-hailing).
	TAUT10	Technology-based traffic management systems (e.g., real-time traffic apps) help me plan my trips.

Variable	Indicator Code	Indicator Statement
	TAUT11	Transport technologies reduce my travel time compared to conventional systems.
	TAUT12	I feel safer when using technology-supported transport (e.g., digital payments, GPS monitoring).
	TAUT13	I believe people in my city are generally ready to accept new transport technologies.
	TAUT14	Socioeconomic conditions (e.g., income, digital literacy) affect readiness to adopt transport technologies.
	TAUT15	Urban Density and spatial structure create the need for smart transport technologies.
Public Attitude Toward Sustainable Mobility (PASM)	PASM1	I have a positive attitude toward using environmentally friendly transport.
	PASM2	I believe sustainable mobility improves the quality of life for citizens.
	PASM3	I feel a moral responsibility to choose environmentally friendly transport modes.
	PASM4	I am highly aware of the environmental impacts of private motor vehicle use.
	PASM5	I feel proud when using sustainable transport because it reflects care for the environment.
	PASM6	I believe sustainable transport contributes to reducing urban air pollution.
	PASM7	I consider electric or eco-friendly public transport better than fossil-fuel private vehicles.
	PASM8	I identify myself as part of a community that cares about the environment through my transport choices.
	PASM9	I believe government support is essential for promoting sustainable transport adoption.
	PASM10	I believe sustainable transport policies help create healthier and more livable cities.
Actual Sustainability Public Transport Use Behavior (ASPT)	ASPT1	I regularly use public transport for my daily activities.
	ASPT2	I use public transport to reduce energy consumption and private vehicle emissions.
	ASPT3	I choose public transport because it is more affordable than private vehicles.
	ASPT4	I use public transport because travel time is relatively efficient.
	ASPT5	I prefer public transport because it is easily accessible from my home or destination.
	ASPT6	I continue using public transport because I feel safe during trips.
	ASPT7	I often use multiple modes (e.g., bus and train) because public transport is well integrated.
Urban Density (UD)	UD1	Population density (people per km ²).
	UD2	Built-up area (% of total urban land).
	UD3	Land use intensity (ratio of building units per hectare).
	UD4	Density of public transport stations and road/rail networks per km ² .